

Monitoring the effects of climate change and topography on vegetation health in Tharparkar, Pakistan

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ABSTRACT

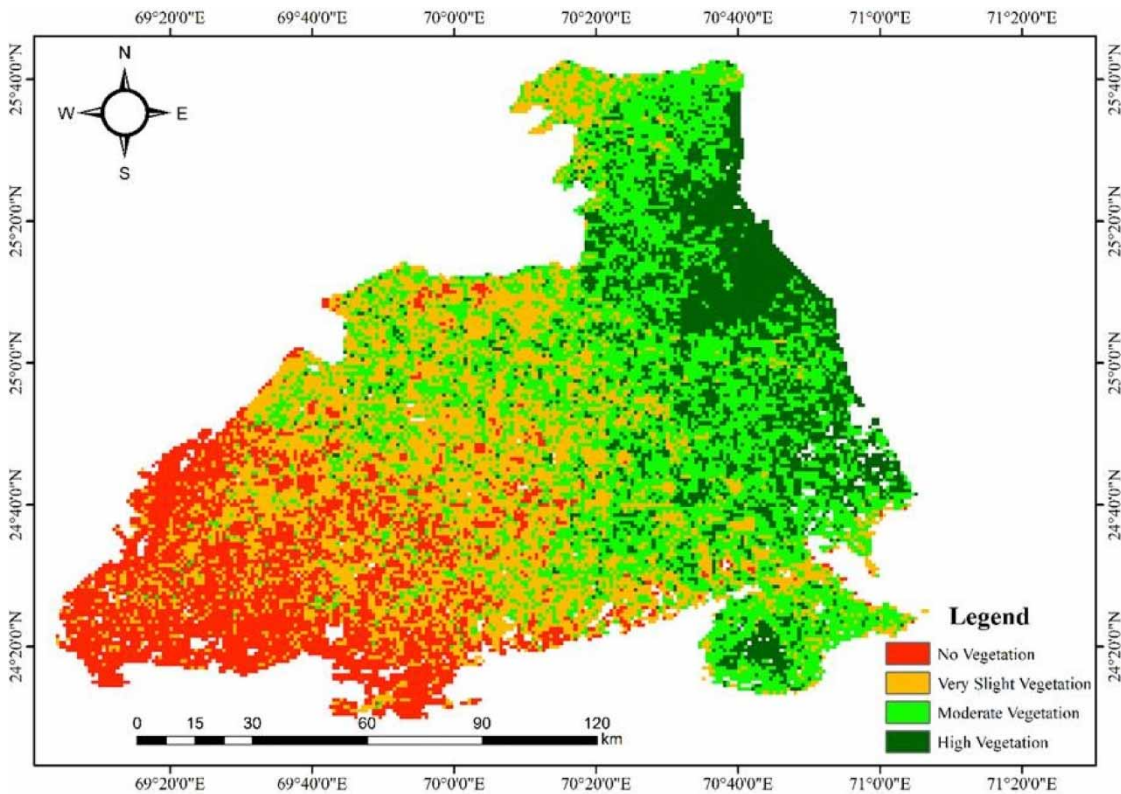
Pakistan's geographic position and socioeconomic profile make it one of the nations that are particularly susceptible to the negative effects of climate change. The Tharparkar district in Pakistan is of particular importance in this regard as it is an arid region with serious environmental issues like drought, desertification, and soil degradation. Therefore, the purpose of this study is to examine how topographic and climatic factors affect vegetation indicators in the Tharparkar. The study utilizes spatiotemporal data spanning over 20 years (2001–2020) collected from the satellites MOD11A2 and MOD13A3. The collected data are processed using a range of tools in ArcGIS 10.4.1, and the impact of topographic and climatic conditions is analyzed based on different vegetation indices, including EVI, NDVI, STVI, OSAVI, and SAVI. The findings reveal that temperature and precipitation, both of which are controlled by topographic features, such as elevation and slope, are the key elements affecting vegetation in Tharparkar. At high elevations, rainfall (>440 mm) and LST (>39 °C) are also high and where the slope is low the density of vegetation indices is high.

Key words: climate change, GIS and remote sensing, multicriteria decision analysis (MCDA), vegetation indices

HIGHLIGHTS

- The findings reveal that temperature and precipitation, both of which are controlled by topographic features, such as elevation and slope, are the key elements affecting vegetation in Tharparkar.
- The study shows that NDVI, EVI, and SAVI rise significantly, whereas vegetation indices such as OSAVI, STVI and LST are adversely impacted by rainfall.

GRAPHICAL ABSTRACT



INTRODUCTION

Climate change is challenging for almost all sectors around the globe. Climate changes have proved to be calamitous for many segments of society, but agriculture can be regarded as a real hard-hit. Climate change is an issue that affects the entire world and has the potential to significantly alter crop yields. There is estimated annual damage to agriculture from rising temperatures. Since the 20th century, human activities such as deforestation and changes in land cover have contributed 1.5 billion metric tons to greenhouse gas (GHG) emissions into the atmosphere, which is one of the main sources of the observed climatic changes (Das & Sharma 2023). These climate changes brought on by human activity also have an impact on the composition of the planet's atmosphere. This enhances the examination of how the climate alters throughout time.

Globally, climate change has in turn caused a decrease in plant cover, which has had significant economic consequences as well as serious losses in biodiversity and the ecosystem (Al-Kindi *et al.* 2023). An ecological element that is particularly vulnerable to climate change is vegetation upon which its impact varies by place and for different plant species. Regions with a delicate ecosystem and climate may experience a greater impact. For instance, GCMs predict warmer winter temperatures, especially in northern latitudes like northern Europe, but also more droughts in some areas due to less precipitation, like southern Europe (AR4 Climate Change 2007: The Physical Science Basis – IPCC n.d.). The loss of urban green space disturbs the ecological balance (Haq 2011). Rapid growth in urbanization also threatens vegetation and open green spaces in urban areas. The reason behind this is that due to urbanization land surface and air temperature increase in such areas (Smith *et al.* 2017). Metropolitan areas like Lahore have less vegetation due to their activities and expansion. Governmental initiatives like Clean and Green Pakistan are established to monitor the vegetation and cover in metropolitan areas. Green surface is the term used to describe the type of vegetation that grows on urban surfaces (Haq 2011).

Climate change is a complex systemic process that causes specific values of spatial variation in vegetated ecosystem responses. The structure and function of ecosystems may be significantly impacted by even small climate changes (Jing *et al.* 2022). Over the past century, a 0.5 °C change in temperature can be observed across different ecological systems including desert ecosystems (McCarty 2001). Major impacts like desertification, loss in productivity, decrease in cultivable land, soil

loss, and water scarcity are caused by climate change in deserts (Goodier 2003). Rainfall patterns are also affected by climate change, which further affects livelihood in the deserts. The main source of livelihood in deserts is livestock keeping, which is dependent on rainfall and is affected by changes in rainfall patterns (Horn *et al.* 2003). Evapotranspiration is also increased by climate change which causes desert warming (Shove 2010).

In order to assist human development, observational data and model simulations offer a fundamental understanding of a variety of hot problems, such as climate change, natural ecosystems, urbanization, and human health (Overpeck *et al.* 2011). However, remote sensing (RS) has steadily risen to prominence as a top study technique in social development studies because it provides multitemporal data and spatially comprehensive images (Yang *et al.* 2013). RS and Geographic Information Systems (GIS) have been widely used to study the role of vegetation trend variation and its relation to the surrounding environment in the last few decades. For instance, Multitemporal images and LULC derived from satellite images along with GIS techniques are used to determine land suitability for agricultural use (AL-Taani *et al.* 2021), to monitor the urban growth and land use change detection (Hegazy & Kaloop 2015), and to identify urban vegetation stress factors (Cârlan *et al.* 2020).

Earth's climate will keep on changing over time. Researchers and policymakers have been searching for multiple ways to track the threat of climate change. Understanding climate change and then forecasting it is a bit difficult for geoscientists as it is determined by a complex set of physical, biological, and chemical interactions. Such type of research requires proper access to data and then managing it properly for further analysis. For this type of analysis, GIS and RS have found wide applications in climate change and adaptation (Eniolorunda 2014). For spatiotemporal analysis, RS imageries have made it possible to analyze the environmental elements and impact of human activities (Lillesand *et al.* 2015). A combined approach of RS and GIS tools can be used to capture and process relevant data for actions according to climate change events (Eniolorunda 2014).

In general, while previous studies on the relationship between vegetation and rainfall have focused solely on drought analysis, this study investigated the relationship between vegetation indices and environmental factors (climate and topography) in the southeastern part of Pakistan's Sindh province for the first time, which could be considered one of the research's innovations. The goal of the study was to analyze how climate change has been affecting Tharparkar's vegetation for the past 20 years by comparing and evaluating data on vegetation. This study aimed to assess the climatic and topographic parameters influencing vegetation cover over 20 years (2001–2020), and investigate the correlation between vegetation indices (Normalized Vegetation Index (NDVI), Stress-Related Vegetation Index (STVI), Optimized Soil-Adjusted Vegetation Index (OSAVI), Soil-Adjusted Vegetation Index (SAVI), and Enhanced Vegetation Index (EVI)) to topographical features using SRTM–Digital Elevation Model (SRTM–DEM), and climatic parameter (CHIRPS) and land surface temperature (LST). Finally the effect, by using a combined approach for all of the parameters using AHP (analytical hierarchy process).

MATERIALS AND METHODS

Study area

Tharparkar is a region of great importance for the study of climate change and its impacts, given its unique environmental and socio-economic characteristics. It lies between 69° 3' 35" E and 71° 7' 47" E longitudes, and between 24° 9' 35" N and 25° 43' 6" N latitudes as shown in Figure 1. Being located in the southeastern part of Pakistan's Sindh province, Tharparkar is one of the least developed areas in the country with a population of over 1.6 million people spread across an area of approximately 19,638 km². The district is characterized by a dry desert landscape with sand dunes and scrubby vegetation that is famous for its desert terrain.

The climate of Tharparkar is defined by hot summers and mild winters, with an average annual temperature of around 27 °C. The majority of the precipitation occurs during the monsoon season, which lasts from July to September, and the region typically receives around 277 mm of rainfall per year, but this varies greatly from year to year (Usman & Nichol 2020). Given the region's dependence on agriculture and livestock rearing, crops such as wheat, cotton, and sorghum are grown in the area. The study area's unique environmental and socioeconomic characteristics make it an important region for studying the impacts of climate change and developing strategies for adaptation and mitigation. The region's vulnerable population and dependence on agriculture and livestock make it particularly susceptible to the effects of climate change, such as prolonged droughts and erratic precipitation patterns. Therefore, understanding the impact of climate change on vegetation cover in Tharparkar is crucial for sustainable development in the region.

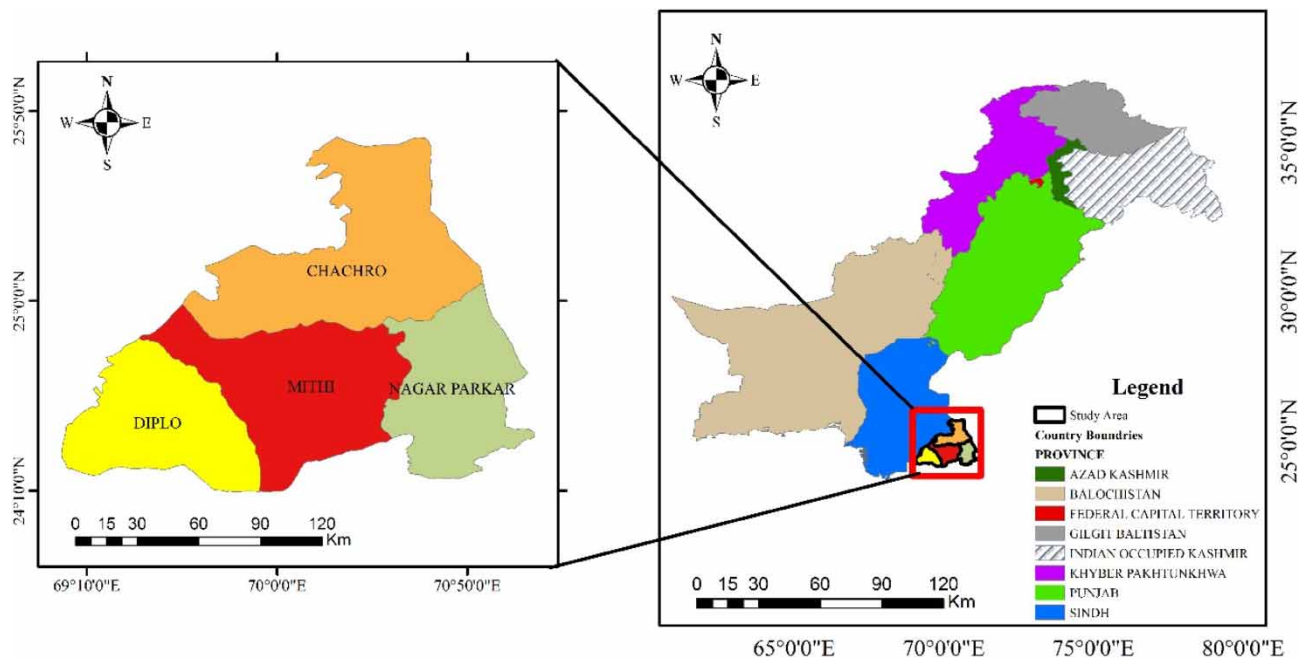


Figure 1 | Location of the study area.

Overall, the study area of Tharparkar presents an excellent opportunity for researchers to analyze and evaluate the impacts of climate change on vegetation indices and to develop strategies to mitigate those impacts. As such, this study's findings can be crucial for policymakers and stakeholders to develop and implement policies that promote sustainable development and adaptation to climate change in Tharparkar and other regions with similar environmental and socioeconomic characteristics.

Data acquisition and processing

Geospatial data for the study area, the Tharparkar region, was collected from various sources, including the US Geological Survey (USGS) and the National Aeronautics and Space Administration (NASA). Table 1 shows the details of all the datasets used in this study. Overall, the dataset includes NDVI, STVI, EVI, SAVI, and OSAVI extracted from MOD13A3 (1,000 m, monthly), LST from MOD11A2 (1,000 m, 8 days), rainfall data through CHIRPS ($0.5^\circ \times 0.5^\circ$), and DIGITAL ELEVATION MODEL (DEM 30×30 m) for slope/elevation aspect extraction. The software used to perform the analysis was ArcGIS 10.4.1 using the multicriteria tools of Spatial Analyst Toolbox.

CHIRPS

The CHIRPS dataset, which combines satellite images with information gathered from on-the-ground meteorological stations, provides a comprehensive source of data on precipitation. It is a useful tool for long-term investigation of precipitation

Table 1 | Types of data, along with their spatial and temporal resolution, sources, and applications, used as inputs for the GIS database

Data	Resolution	Resource	Time	Source
MOD13A3	Monthly 1 km resolution	Vegetation indices (NDVI, STVI, EVI, OSAVI)	01-01-2001 to 31-12-2020	https://earthexplorer.usgs.gov/
MOD11A2	Daily 1 km resolution	Land Surface Temperature (LST-day and LST-night)	1-1-2001 to 31-12-2020	https://earthexplorer.usgs.gov/
DEM	$30 \text{ m} \times 30 \text{ m}$	Elevation aspect	1-1-2001 to 31-12-2020	http://dwtkns.com/srtm30m/
CHIRPS	$0.05^\circ \times 0.05^\circ$	Precipitation data	1-1-2001 to 31-12-2020	http://app.climateeng.org/climateengine

patterns since it offers a high level of precision and has been around since 1981 (Hsu *et al.* 2021). The dataset, which contains data on rainfall distribution and amount, can be used to examine historical changes in precipitation patterns. The algorithm used by CHIRPS is based on a very precise grid of $0.05^\circ \times 0.05^\circ$ with variables of stations as follows:

Time: (Julian day) order = (1 January 2001) to (31 December 2020)

No formula was used in this calculation. Shapefile of Tharparkar consisting of points (stations) which were taken from Google Earth were added to ArcGIS to generate precipitation raster. Table 2 shows the longitudes and latitudes of all the stations.

Land surface temperature

The temperature of the land surface, which includes the soil and vegetation, is referred to as LST. As a reflection of the equilibrium of energy and water exchange between the land surface and the atmosphere, it is a significant factor in the Earth's climate system.

The amount of solar radiation absorbed, the moisture level of the soil, and the kind and density of vegetation cover are only a few of the variables that have an impact on LST. The timing and rate of plant growth, as well as the frequency and intensity of heat waves and droughts, can all be significantly impacted by changes in LST. The formula used to calculate LST is shown in Equation (1) as follows (Dong *et al.* 2015):

$$\text{LST} = \text{DN} * 0.02 - 273.15 \quad (1)$$

where LST is the land surface temperature and DN is the digital number value.

Enhanced Vegetation Index

The EVI from Landsat is used to describe vegetation greenness as it enhances the NDVI's quality. Twenty years of data were acquired for this study. EVI typically ranges from -1 to 1 , and in the case of healthy vegetation, it fluctuates between 0.2 and 0.8 . The general formula used to calculate EVI is shown in Equation (2) (Zhang *et al.* 2003; Ihuoma & Madramootoo 2019):

$$\text{EVI} = \frac{2.5 * (\rho \text{ NIR} * \rho R)}{(\rho \text{ NIR} + C1 * \rho R - C2 * \rho B + L)} \quad (2)$$

Here, L is the canopy background adjustment = 1 , $C1$ is coefficient used for aerosol resistance = 6 , $C2$ is also coefficient used for aerosol resistance = 7.5 , ρR = reflectance of red band, and $\rho \text{ NIR}$ = reflectance of near-infra-red band.

There was no need for a calculation since the EVI was already available in MODIS data bundle.

The NDVI

The NDVI is a recognized method for obtaining healthy vegetation that quantifies the difference between red and near-infrared light. The NDVI's range is from -1 to $+1$, and moderate NDVI values may be caused by sparse vegetation, such as shrubs and grasslands or senescent crops (approximately 0.2 – 0.5). The general formula used to calculate EVI is shown in Equation

Table 2 | Longitudes and latitudes of stations

ID	Stations	Latitude	Longitude
1.	Diplo	24.5124	69.5738
2.	Haronabad	29.31297	73.14086
3.	Vinjanari	24.50312	70.05925
4.	Islamkot	24.7014	70.1783
5.	Nagarparkar	24.3572	70.7555
6.	Vandhio	24.2397	70.6439

(3) (Huete *et al.* 2010):

$$NDVI = \frac{(\rho \text{ NIR} - \rho R)}{(\rho \text{ NIR} + \rho R)} \quad (3)$$

where ρR = reflectance of red band, and $\rho \text{ NIR}$ = reflectance of near-infra-red band.

The NDVI data were also already available in the MODIS product when the data were downloaded, thus there was no need to calculate it numerically using the raster calculator in ArcMap software.

The STVI

The STVI is a vegetation indicator that is used to measure plant stress brought on by a variety of condition including environmental pollution, disease, pests, and drought. In dry and semi-arid areas, the indicator is especially helpful in tracking vegetation stress. Equation (4) is used to determine STVI for each month across the 20-year study period (Basso *et al.* 2004):

$$SI = \frac{(\rho \text{ MIR} * \rho R)}{(\rho \text{ NIR})} \quad (4)$$

where ρR = reflectance of red band, $\rho \text{ MIR}$ = reflectance of mid-infra-red band, and $\rho \text{ NIR}$ = reflectance of near-infra-red band.

The indicator is based on the idea that compared to stressed vegetation, healthy vegetation reflects more near-infra-red light and less mid-infra-red and red radiation. Hence, STVI levels near 1 denote healthy vegetation, while STVI values near -1 denote stressed vegetation.

Soil-Adjusted Vegetation Index

SAVI is a modification of the NDVI that is designed to reduce the influence of soil reflectance on vegetation indices. The SAVI formula is shown in Equation (5) as follows (Farg *et al.* 2012):

$$SAVI = \frac{(\rho \text{ NIR} - \rho R)}{(\rho \text{ NIR} + \rho R + L)} * (1 + L) \quad (5)$$

Here, L is a soil adjustment factor that is typically set to 0.5, ρR = reflectance of red band, and $\rho \text{ NIR}$ = reflectance of near-infra-red band.

SAVI is especially valuable in arid and semi-arid areas where there is little plant cover and soil reflectance can significantly affect the values of the overall vegetation index.

The OSAVI

OSAVI is essentially based on SAVI. This index utilizes the L factor value of 0.16. SAVI functions best in regions where the soil can be seen through subsurface drip irrigation (SDI)ing. The brightness of the soil does not affect it. This index is effective in areas with little vegetation and apparent soil. The formula for OSAVI is a little modified version of SAVI shown by Equation (6) (Ren *et al.* 2018):

$$SAVI = \frac{(\rho \text{ NIR} - \rho R)}{(\rho \text{ NIR} - \rho R + 0.16)} \quad (6)$$

Here, ρR = reflectance of the red band, and $\rho \text{ NIR}$ = reflectance of the near-infra-red band.

Analytical hierarchy process

The methodology involved the use of the AHP as a multicriteria decision-making method to determine the relative importance of different criteria and select the best option among a set of alternatives. Data for all factors involved in the decision-making process were gathered from various websites.

The first step involved converting the vector data, such as CHIRPS data, into raster format using the rasterization tool in ArcMap 10.4.1. Vegetation indices were then derived from MODIS images using various formulas from Equations (1)–(6).

After calculating the LST and Vegetation Indices, the resulting raster data were reclassified into different categories ranging from poor to excellent vegetation cover based on the standard deviation of the data. Reclassification allowed for visual analysis of the patterns and changes in vegetation cover and temperature in the study area.

Next, the AHP tool was employed, which involved comparing each pair of all the criterion rasters and assigning relative importance to each criterion. A comparison table marked from 1 to 9 scale was used, where 1 represented equal importance, and 9 represented extreme importance.

Once the pairwise comparisons were completed, the criterion weights were normalized in the comparison matrix to ensure that the weights added up to one and each criterion's relative importance was accurately reflected. The flowchart for the methodology employed in the study to carry out the analysis is shown in Figure 2.

Table 3 is used to show the relative importance of each criterion in determining vegetation cover suitability using rate, Normalized Rating Index (NRI) and weight index based on Multi-Criteria Decision Analysis (MCDA). Further, the final raster map resulting from the integration of all the factors shows results in the form of the vegetation map with four categories: no vegetation, very slight, moderate, and high vegetation. This final raster map is a visual representation of the vegetation cover in the study area based on different indices, LST, slope, elevation, and rainfall.

To determine the most significant contributing factors, a correlation analysis was performed between the satellite-based and ground-based data. The correlation was computed among rainfall, LST, EVI, NDVI, STVI, OSAVI, and SAVI, and the results indicate a significant correlation at the 0.01 level and 0.05 level (two-tailed). The use of correlation analysis helps in identifying the key factors that are responsible for the vegetation cover in the study area and can be used for future studies and policymaking.

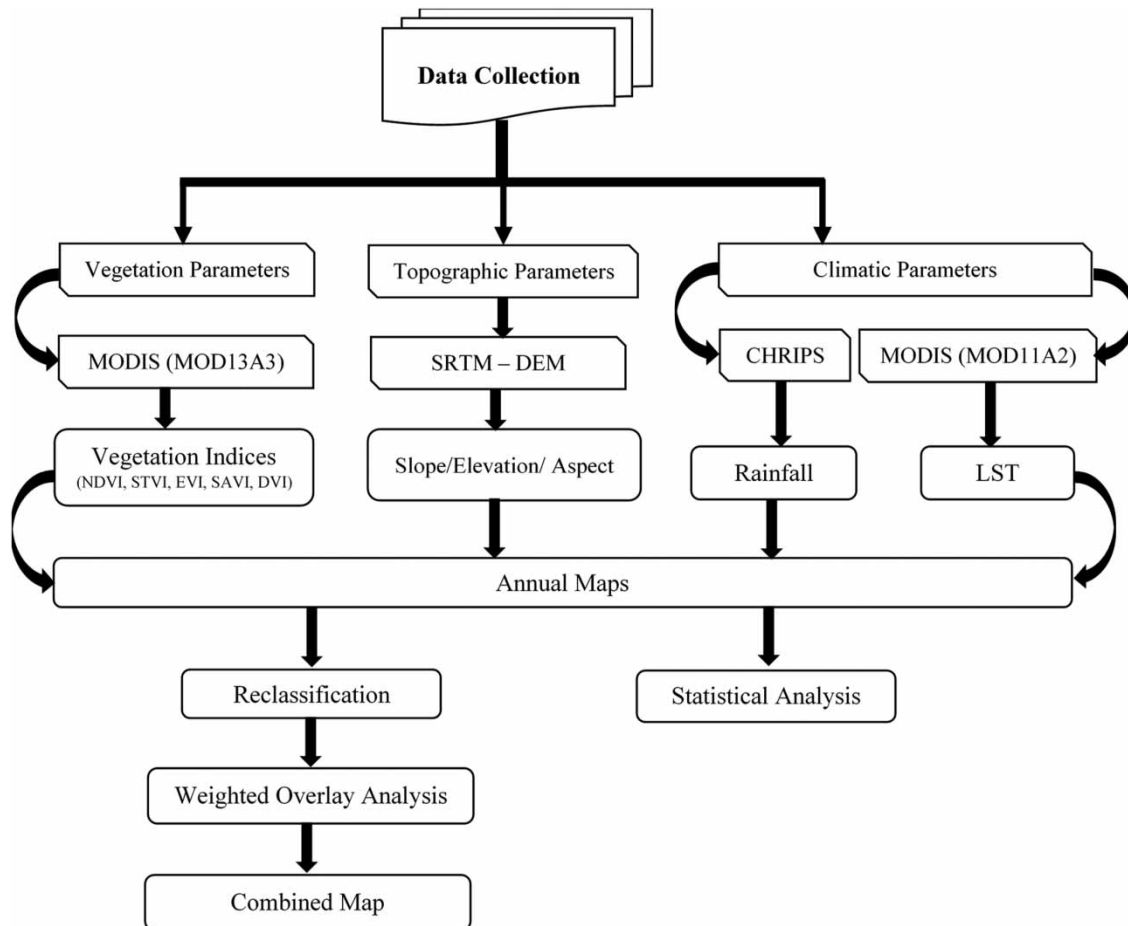


Figure 2 | Methodology flowchart.

Table 3 | Rate, normalized rating index and weight index based on MCDA

Parameter	Category/Class	Descriptive level	Rate (R)	Normalized Rating Index (NRI)	Weight	Normalized Weight Index (NWI)
Rainfall (mm)	≤340	Very Low	1	0.07	9	0.20
	340–390	Low	2	0.13		
	390–440	Moderate	3	0.20		
	440–490	High	4	0.27		
	>490	Very High	5	0.33		
LST	<35	Very Low	1	0.07	8	0.18
	35–37	Low	2	0.13		
	37–39	Moderate	3	0.20		
	39–41	High	4	0.27		
	>41	Very High	5	0.33		
Slope	>13	Very Low	1	0.07	7	0.16
	7.0–13	Low	2	0.13		
	4.0–7.0	Moderate	3	0.20		
	1.5–4.0	High	4	0.27		
	≤1.5	Very High	5	0.33		
Elevation	>130	Very Low	1	0.07	6	0.13
	100–130	Low	2	0.13		
	70–100	Moderate	3	0.20		
	50–70	High	4	0.27		
	≥50	Very High	5	0.33		
EVI	<0	Very Low	1	0.07	5	0.11
	0–0.1	Low	2	0.13		
	0.1–0.2	Moderate	3	0.20		
	0.2–0.4	High	4	0.27		
	> 0.4	Very High	5	0.33		
NDVI	<0	Very Low	1	0.07	4	0.09
	0–0.1	Low	2	0.13		
	0.1–0.2	Moderate	3	0.20		
	0.2–0.4	High	4	0.27		
	>0.4	Very High	5	0.33		
STVI	>0.3	Very Low	1	0.07	3	0.07
	0.2–0.3	Low	2	0.13		
	0.1–0.2	Moderate	3	0.20		
	0–0.1	High	4	0.27		
	≤0	Very High	5	0.33		
OSAVI	≤0	Very Low	1	0.07	2	0.04
	0–0.1	Low	2	0.13		
	0.1–0.2	Moderate	3	0.20		
	0.2–0.3	High	4	0.27		
	> 0.3	Very High	5	0.33		
SAVI	≥0	Very Low	1	0.07	1	0.02
	0–0.1	Low	2	0.13		
	0.1–0.2	Moderate	3	0.20		
	0.2–0.3	High	4	0.27		
	>0.3	Very High	5	0.33		

Pearson's correlation coefficient analysis

Pearson's correlation coefficient is a statistical test that measures the mathematical relationship between two continuous variables. It provides details about the strength and direction of the association. This study utilized Pearson's correlation coefficient analysis to check the existence of a linear relationship between input variables. A strong association is indicated by a high correlation, while a weak relationship is shown by a low correlation. The magnitude of 'r' represents the strength of the association, whereas the sign of 'r' denotes the direction of the association. A correlation near 0 indicates no linear

relationship between two continuous variables. Pearson's correlation coefficient analysis is frequently used to assess the strength of the relationship between two quantitative variables when statistical data are available (Sedgwick 2012; Sibley *et al.* 2014).

Table 4 displays the correlation chart for all the components, providing a comprehensive overview of the relationship between different factors and their impact on vegetation cover. This analysis evaluated the significance of these parameters in comparison to each other. Mainly two trends are observed, one related to rainfall and the other to temperature, both of which are influenced by topographic features such as slope and elevation.

RESULTS AND DISCUSSIONS

Correlation analysis and interpretation

The study examined the relationships between different factors in the study area, including rainfall, LST, elevation, and vegetation indices. Generally, two patterns are seen from the correlation analysis of these factors: one from the standpoint of rainfall and the other from that of temperature topographic features (slope and elevation) influence these climate parameters.

Effect of rainfall and temperature

Rainfall, LST, EVI, NDVI, STVI, OSAVI, and SAVI showed a correlation significant at the 0.01 level (two-tailed). Overall, rainfall causes the vegetation indices (NDVI, EVI, SAVI) to increase positively and affect (LST, OSAVI, and STVI) negatively because the temperature of the area drops, and due to that stress level decreases (Castillo *et al.* 2007). The NDVI and rainfall association is quite substantial and favorable with a correlation coefficient of 0.495 ($p = 0.01$). It shares a similar relationship with both EVI ($r = 0.533$, $p < 0.01$) and SAVI ($r = 0.345$, $p < 0.01$) as well. However, there is not much significant link between LST and STVI, with a correlation coefficient of -0.127 ($p = 0.334$). When comparing LST to other indices, the results will be ambiguous; rainfall will have a negative effect on LST, but NDVI, EVI, and SAVI show significant values of 0.05 (Nwilo *et al.* 2021). LST and STVI have a bad correlation. The results will be contrary when comparing LST to other indices; there will be a negative response to rainfall, with NDVI, EVI, and SAVI exhibiting significant values of 0.05.

An individual parameter were reclassified into five classes i.e. very low, low, moderate, high and very high as shown in Figure 3. It is visually interpreted from the reclassified raster images of the study area that Nagar Parkar and Chachro receive moderate to high rainfall, while Mithi and Diplo experience relatively lower rainfall levels (Figure 3). Land surface temperatures were high to moderate in all districts, ranging from 37 to 41 °C. The study area appeared to be generally low-lying with a nearly uniform slope, but with varying elevations from Chachro (over 130 km high) to Diplo (less than 50 km high). Vegetation indices were generally moderate in areas within Mithi and some areas of Diplo, while Nagar Parkar and Chachro showed relatively lesser vegetation illustrating that areas with higher elevation and LST tended to have lower vegetation NDVI and SAVI values. Overall, the statistical analysis using the Pearson Correlation and the visual interpretations from reclassified raster corroborate the results of each other and give insights into the dynamics of the research region.

Table 4 | Correlation among rainfall, NDVI, EVI, SAVI, LST, STVI, and OSAVI of the study area

		Rainfall	NDVI	EVI	SAVI	LST	STVI	OSAVI
Rainfall	Pearson Correlation	1	0.495**	0.533**	0.345**	-0.252	-0.127	-0.530**
NDVI	Pearson Correlation	0.495**	1	0.993**	0.554**	-0.483**	-0.640**	-0.789**
EVI	Pearson Correlation	0.533**	0.993**	1	0.533**	-0.455**	-0.620**	-0.771**
SAVI	Pearson Correlation	0.345**	0.554**	0.533**	1	-0.305*	-0.516**	-0.519**
LST	Pearson Correlation	-0.252	-0.483**	-0.455**	-0.305*	1	0.256*	0.472**
STVI	Pearson Correlation	-0.127	-0.640**	-0.620**	-0.516**	0.256*	1	0.588**
OSAVI	Pearson Correlation	-0.530**	-0.789**	-0.771**	-0.519**	0.472**	0.588**	1

**Correlation is significant at the 0.01 level (two-tailed).

*Correlation is significant at the 0.05 level (two-tailed).

Temporal trends of climate and vegetation patterns

To comprehend the effect of rainfall and temperature on vegetation, it is necessary to understand its spatial and temporal dynamics (Krogulec & Zabocki 2015). Graphs are used to support the relationships discovered by correlation analysis and classification maps. Figure 4(a)–4(c) shows the temporal trend of indices, precipitation and LST of the study area. According to the graph (Figure 4(a)), the vegetation indices SAVI, NDVI, and EVI show a steadily rising trend. This indicates that vegetation and rainfall are directly related, as an increase in rainfall (Figure 4(b)) over time corresponds with an increase in vegetation. On the other hand OSAVI and STVI indices showed a decreasing trend indicating the decrease in vegetation

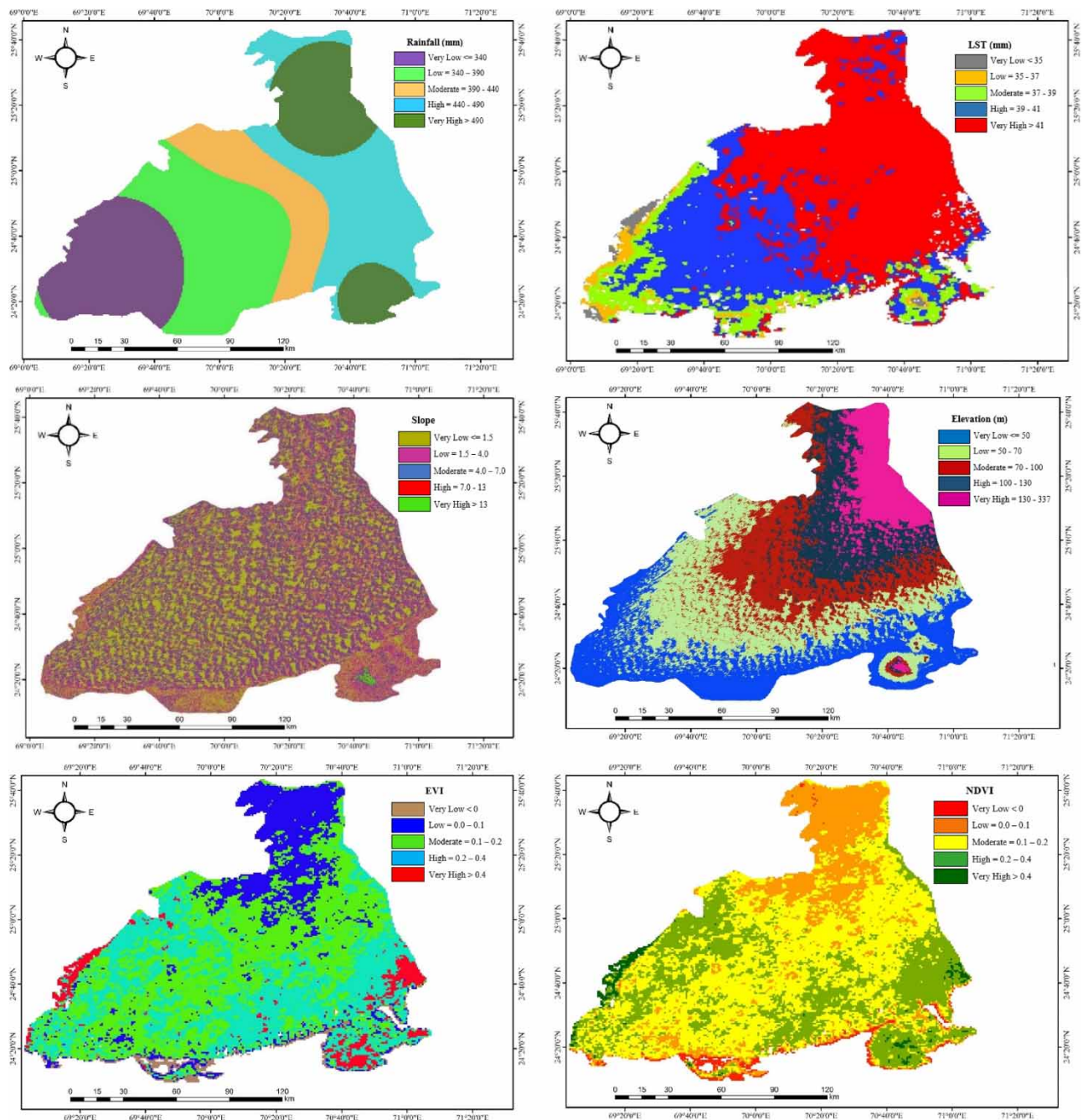


Figure 3 | Reclassified raster for each factor. (continued)

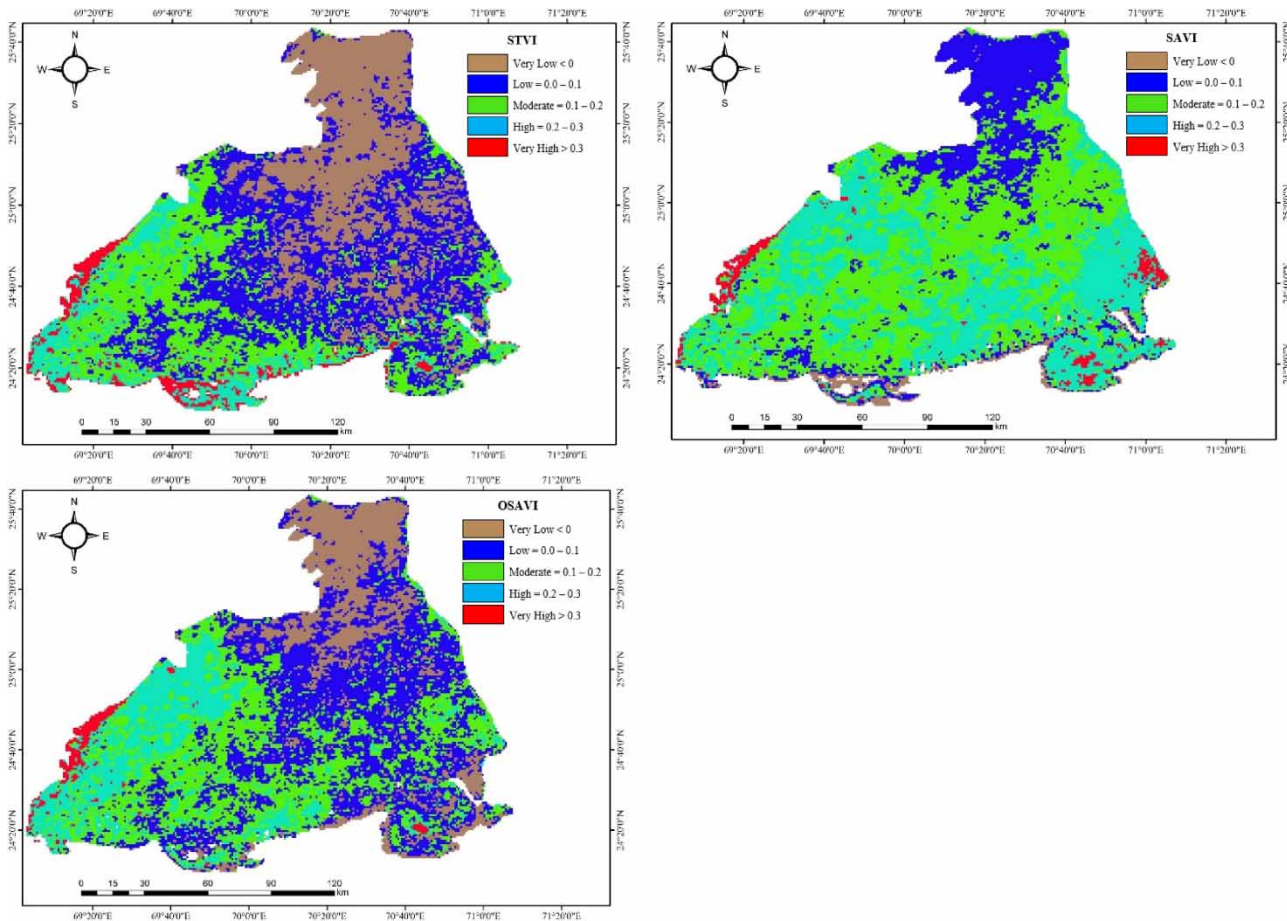


Figure 3 | Continued.

based on certain conditions. In addition, LST shows a peak in 2001 that indicates the increase in temperature during this timeline, the peak followed a steady fall through 2011 indicating the cooling trend. After 2011 there is a substantial increase in the trend up to 2022 which indicates the sudden rise in the temperature shown in [Figure 4\(c\)](#). So according to the temperature trend negative correlation is observed between LST and STVI which means that the decrease in vegetation density is associated with higher temperature other than that significant positive correlation between precipitation and various vegetation indices such as OSAVI, NDVI, and EVI is also observed. These trends may vary seasonally causing different changes in vegetation density in different times of the year.

Spatial mapping of vegetation density using AHP

AHP, assisted in producing a classified vegetation map of the study area, as shown in [Figure 5](#), based on vegetation indices, temperature, precipitation (climatic variables), slope and elevation (topographic factors), which have a big impact on EVI, NDVI, STVI, OSAVI, and SAVI (vegetation indices). The vegetation density is comparatively high for regions like Khensar, Nagarparkar, Vandhio, and Haronabad, which are areas with moderate elevation, rainfall, and lower LST ([Mukwada & Manatsa 2018](#)). Islamkot, Vinjanria, and Mithi, in contrast, have witnessed a moderate to slight decrease in trend as a result of the inverse patterns of rainfall and topographic variables in these regions. Diplo region, which has the lowest elevation and rainfall, has no significant vegetation at all.

In general, temperature and rainfall served as the two key indicators in a study that shows the trends in this area. The findings suggested that, as time passed, rainfall and temperature variations had an impact on vegetation development. The growth of vegetation was observed to be stressed during times of low rainfall and high temperatures ([Cotrina Cabello et al. 2023](#)).

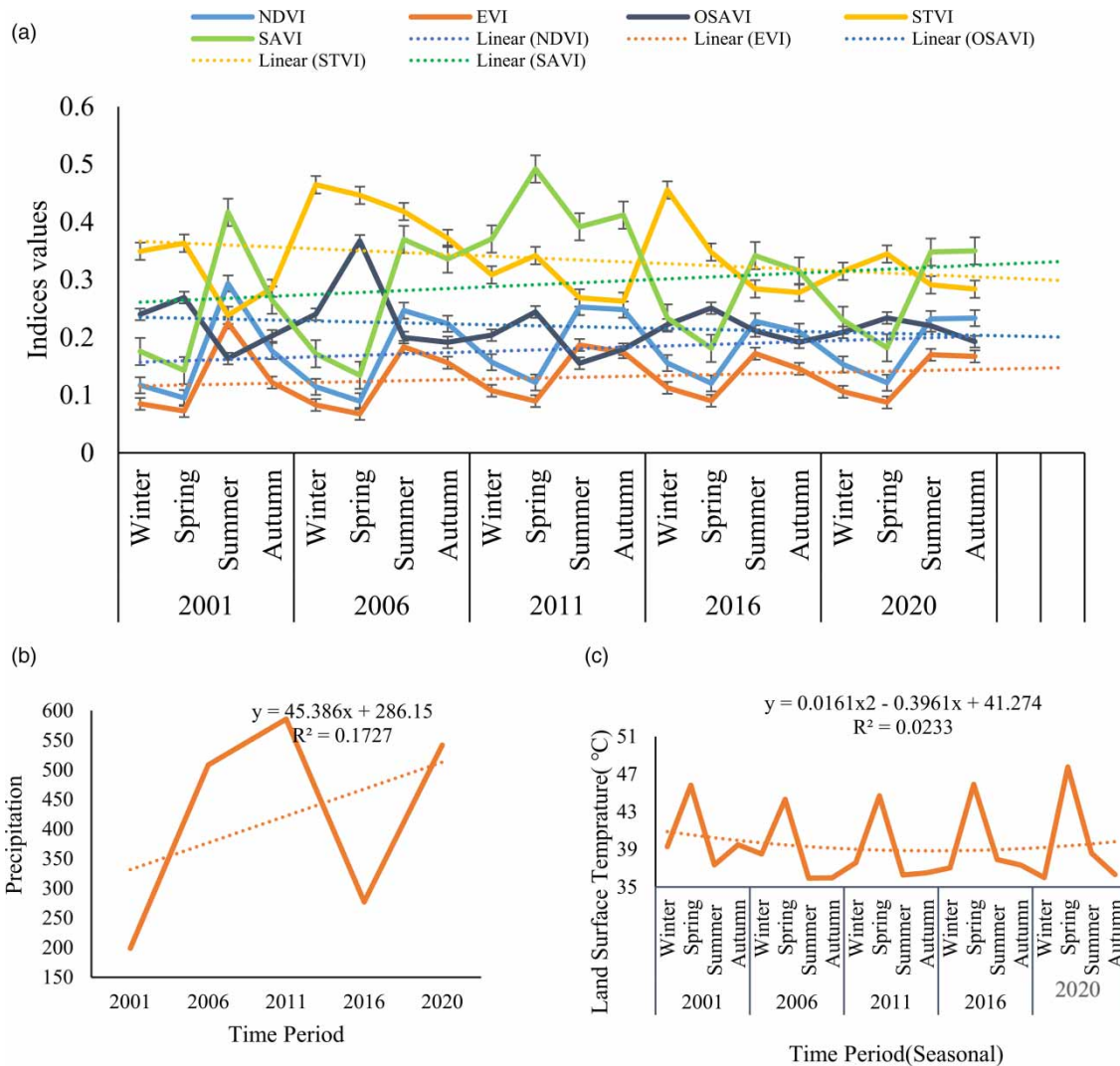


Figure 4 | (a) Vegetation indices, (b) rainfall and (c) land surface temperature trends.

When there is more rainfall, stress and temperature are lower, resulting in increased vegetation density, and vice versa (Joshi et al. 2019), similar effects are seen in our study that rainfall and temperature both had similar effect on NDVI and SAVI. Both vegetation indices had direct relation with the rainfall and inverse relation with temperature, similar effect was observed for EVI as well. Future research could incorporate additional criteria and factors for site suitability analysis to improve the accuracy of the results. Nonetheless, the results of this study provide valuable insights for decision-makers in the energy industry and can aid in the development of sustainable energy solutions.

CONCLUSION

This research underscores the vital link between climatic fluctuations and vegetation in Tharparkar, Pakistan. Through an examination of climatic and topographic factors alongside vegetation indices, it becomes evident that each factor influences the others in diverse ways. Specifically, our analysis reveals that temperature and precipitation (climatic factors), as well as slope and elevation (topographic factors), exert significant effects on EVI, NDVI, STVI, OSAVI, and SAVI (vegetation indices) in the Tharparkar region.

Moreover, our findings indicate that elevated areas having elevation ≤ 130 m experience heightened LST ≤ 41 °C and rainfall ≤ 490 mm exhibit moderate to high levels of vegetation cover. Conversely, regions with lower elevations having elevation

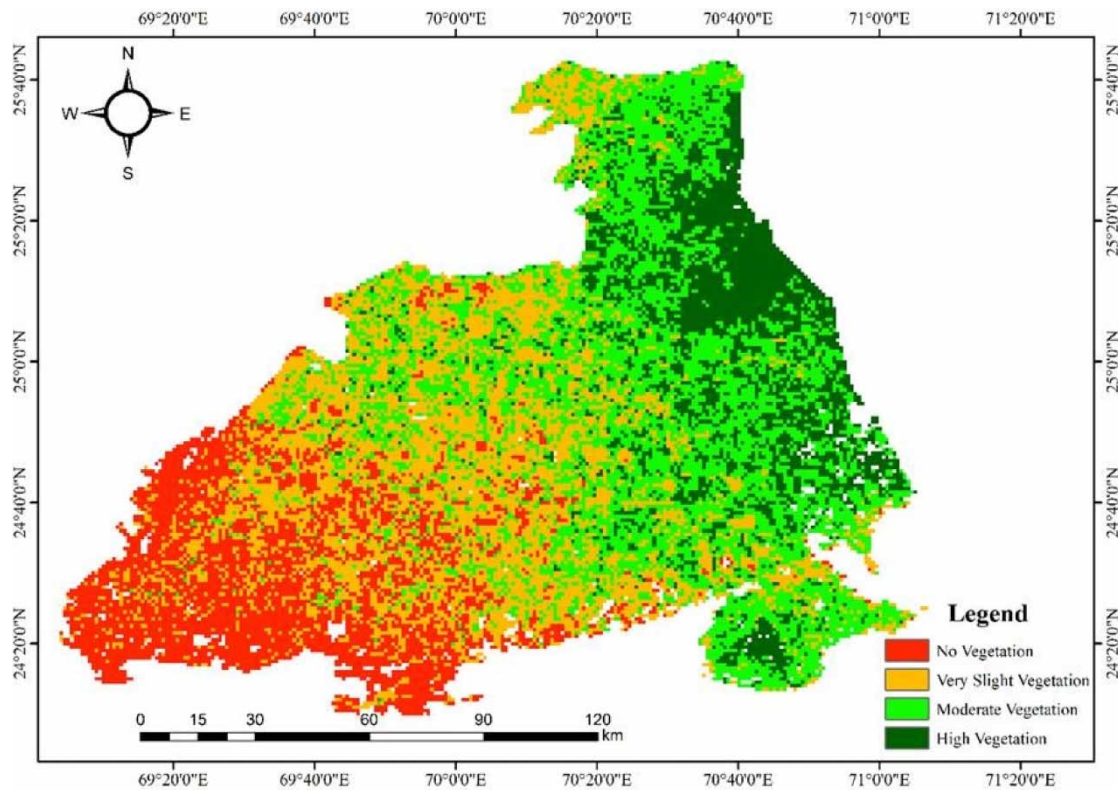


Figure 5 | Combined vegetation map.

≥ 50 m having low rainfall ≥ 340 mm placing stress on vegetation and leading to reduced vegetation and canopy cover. Additionally, our study underscores the pivotal role of STVI in gauging low vegetation stress in high-elevation and high-temperature and high-rainfall zones, with stress levels gradually increasing as elevation and rainfall decreases. The combined map confirms that the upper part of the study area, characterized by high elevation and significant rainfall, is classified as having moderate to high vegetation cover, whereas the lower part, with lower elevation and less rainfall, exhibits the opposite.

Overall, this research underscores the imperative of investigating and addressing climate change in historically overlooked regions like Tharparkar. Our findings elucidate the significant impact of climatic factors on vegetation cover in this area, contributing to its susceptibility to drought and harsh environmental conditions. This study serves as a clarion call to prioritize measures for climate change mitigation and adaptation to safeguard the ecological health and resilience of Tharparkar and similar regions. Utilizing GIS technology, this approach identifies regions experiencing extreme temperatures, analyzes the impact of global warming, and evaluates changes in terrestrial cover due to climate change. It facilitates the identification of areas most vulnerable to climate change, aiding in better natural resource management. Additionally, advocating for adequate government funding and resources is crucial to support research in this field, which is essential for understanding and mitigating climate change's effects on vegetation.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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